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3D Displacement From Shaded Relief Feature Offsets: Multi-Decadal Deceleration and Secondary Mini-Sliding at the Slumgullion Landslide, Colorado, USA

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Abstract The Slumgullion landslide in Colorado, USA, is a well-known natural laboratory for studying mass-wasting processes, owing to its ~4 km length and sustained movement averaging a couple of centimeters per day over the past ~300 years. However, key questions remain about how its margins and internal boundaries evolve, whether the headscarp still supplies material, and whether motion is accelerating or slowing over decadal timescales. Here we conducted offset tracking on light detection and ranging (LiDAR)-derived shaded relief images to resolve three-dimensional landslide displacements, revealing previously unrecognized secondary mini-slides close to Slumgullion's margins. Comparing our results to those from previous displacement studies revealed the landslide's long-term deceleration of motion since 1939. The secondary failures near the landslide margins and at the headscarp confirm continued material supply to the active slide. The Slumgullion landslide remains structurally dynamic with the maximum shear strain rates at ~1 yr^{-1}. Despite this activity, toe advance has decelerated by ~36% since the 1990s. The approach provides a transferable framework for mapping three-dimensional deformation using high-resolution shaded relief images for landslides and other geohazards.

Plain Language Summary The Slumgullion landslide in Colorado is one of the world's best-known natural sites for studying how landslides move, because it has been active for about 300 years. Even so, important questions remain about whether its edges and internal zones are changing, whether the upper portion is still feeding new material into the slide, and whether the landslide is speeding up or slowing down over time. In this study, we measured landslide motion by tracking offsets on the shaded relief images made from high-resolution light detection and ranging (LiDAR) topography to reconstruct three-dimensional landslide movement. The method revealed several small secondary slides near the landslide margins and at the headscarp that had not been recognized before. Our results show that the Slumgullion landslide is still structurally active, with new material continuing to enter the system. At the same time, the overall movement has slowed: the toe of the landslide has decelerated by about 36% since the 1990s. These findings improve our understanding of how large landslides evolve over decades and show that this method could also be used to study other landslides and geologic hazards.

Key Points:

- Offset tracking on shaded relief images provides 3D landslide movements, revealing multiple mini-slides close to the landslide margins
- Slumgullion landslide's toe advance decelerated by ~36% since 1990s
- Slumgullion landslide exhibits high internal strain rates of ~0.1 yr^{-1}, two orders of magnitude larger than other slow-moving landslides

Supporting Information:

Supporting Information may be found in the online version of this article.

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1. Introduction

The Slumgullion landslide is a translational earth slide in the San Juan Mountains, Hinsdale County, southwestern Colorado (Figure 1a). The landslide consists of hydrothermally altered volcanic clay, silt, and rock fragments ranging from rhyolite to basalt (Varnes & Savage, 1996) within a surrounding geologic setting of heterogeneous Tertiary volcanic rocks and Quaternary surficial deposits (Figure S1 in Supporting Information S1). The landslide has experienced two primary active episodes. A large failure occurred about 700 years ago, damming the Lake Fork of the Gunnison River and creating Lake San Cristobal, the second largest natural lake in Colorado. The

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currently active part of the landslide has persisted for the past ∼300 years within the old landslide deposits, extending downslope for about 3.9 km from the original headscarp, composed primarily of Tertiary volcanic rocks, and terminating above present-day County Highway 149. Like many slow-moving landslides, Slumgullion exhibits a seasonal acceleration driven by hydroclimatic forcings (e.g., Shi et al., 2021; Zhou et al., 2026).

This enigmatic mass-wasting process, lasting for hundreds of years, has drawn the attention of successive generations of scientists worldwide since the 1800s (e.g., Endlich, 1876). In the early 1990s, scientists from the U.S. Geological Survey (USGS) and the Istituto di Ricerca per la Protezione Idrogeologica (IRPI) of the Italian National Research Council conducted an in-depth field survey of the Slumgullion landslide, delineating the landslide's external margins and internal boundaries, as well as representative geomorphological features (Fleming et al., 1999). Although the kinematic units mapped during the 1991 field survey did not capture every displacement discontinuity or subtle topographic break, the broader segmentation among the head, stretching zone, neck, main body, and a partly detached toe remains distinct in the landslide's present-day appearance (e.g., Hu, Bürgmann, Schulz, & Fielding, 2020). A wide array of remote sensing tools including stereo aerial photography, geodetic sensors such as traditional control surveys, global navigation satellite system (GNSS)/global positioning system (GPS), extensometers, inclinometers, light detection and ranging (LiDAR), and synthetic aperture radar (SAR), and geophysical sensors such as seismic arrays, have been deployed across the landslide mass to monitor its behaviors. For example, as early as the 1960s, traditional control surveys documented average velocities of ∼1.6 cm day^{-1} at the landslide neck and ∼0.4 cm day^{-1} at the toe (Jackson et al., 1996). In the 1990s, USGS researchers deployed creepmeters, a portable seismic network, and GPS on the Slumgullion landslide. Data from 19 GPS stations deployed across the landslide indicate that velocity peaked in the midsection (Coe et al., 2000) (Figure 1b). Creepmeter records showed pronounced seasonal variability, with an average rate of ∼1.5 cm day^{-1} along the south-bounding strike-slip fault (Savage & Fleming, 1996). The seismic network detected slidequakes, characterized by short-duration signals that were spatially and temporally clustered (Gomberg et al., 1996). In June 1993, GPS surveys measured daily velocities of ∼1.2–1.5 cm day^{-1} in the center of the active landslide during a 4-day campaign (Jackson et al., 1996). Coe et al. (2009) used stereo aerial photographs from October 1939, October 1940, and July 2000 to quantify topographic change over a ∼60-year interval. Coe (2012) forecasted a decreasing landslide movement by referring to the moisture balance drought index. Schulz, Kean, and Wang (2009) reported that daily acceleration episodes coincided with atmospheric low-pressure conditions based on extensometer and piezometer observations. Gomberg et al. (2011) combined a 29-element prism array surveyed with a robotic theodolite and an 88-station seismic network to demonstrate largely aseismic deformation along the basal surface. More recently, advances in ground-based, airborne and spaceborne SAR have enabled detailed mapping of Slumgullion surface displacements using offset tracking and interferometric SAR (InSAR), including studies using Ingegneria dei Sistemi IBIS-L (ground-based), NASA/JPL UAVSAR (airborne), Artemis SlimSAR (airborne), Italian COSMO-SkyMed (spaceborne), and European Space Agency's Copernicus Sentinel-1 (spaceborne) (e.g., Cao et al., 2017; Delbridge et al., 2016; Hu, Bürgmann, Schulz, & Fielding, 2020; Milillo et al., 2014; Schulz et al., 2017). Because high displacement gradients complicate phase unwrapping, Molan and Lohman (2023) unwrapped UAVSAR interferograms using a prior reference displacement field and evaluated results against offset-tracking estimates. One of the most comprehensive spatiotemporal displacement data sets to date was produced by jointly exploiting Sentinel-1 and UAVSAR observations with combined InSAR and offset-tracking methods (Hu, Bürgmann, Schulz, & Fielding, 2020). More recently, Structure-from-Motion (SfM) photogrammetry was used to process georeferenced images acquired on 3 June 2024 during four UAV sorties, generating a point-cloud product and a digital surface model (Hoch et al., 2025).

Despite numerous studies, important questions remain to be answered: Is the landslide head stable, or is it still providing loose materials to the system? Are the landslide movements speeding up or slowing down? How does the strain rate at Slumgullion compare with those of other slow-moving landslides and fault systems? Answering these questions is beneficial for understanding the landslide's stability, evolution, deformation mechanisms, and potential hazards. It requires long-term, spatially continuous, high-resolution displacement measurements. In this study, we mapped landslide movements using multi-temporal LiDAR digital elevation model (DEM) at much finer spatial scales than in previous work, revealing new slide features and a long-term deceleration of landslide movement since the 1930s.

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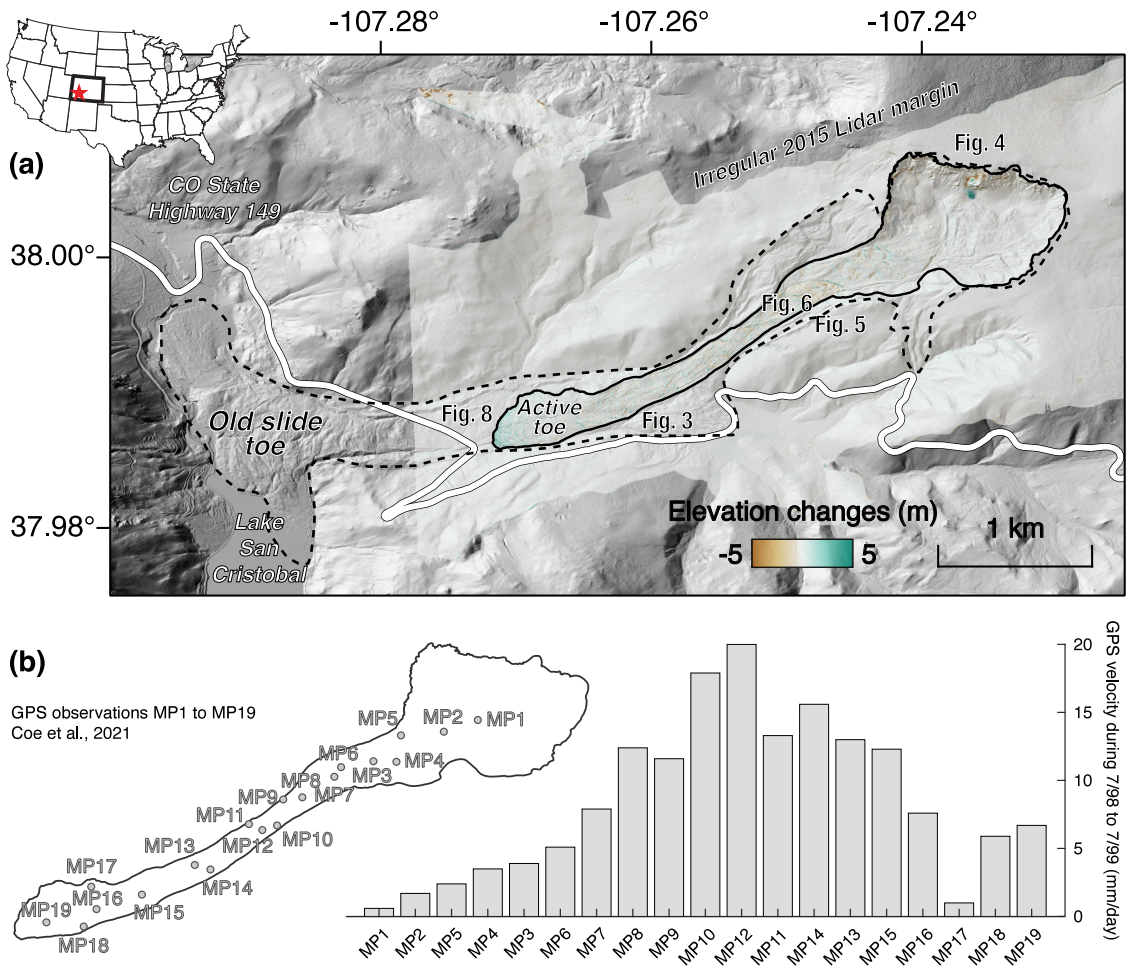


Figure 1. Slumgullion landslide, Colorado, USA. (a) Light detection and ranging digital elevation model (LiDAR DEM) difference between 2015 and 2019 overlaid on a shaded relief image of the 2019 LiDAR DEM. The 2015 DEM is downloaded from OpenTopography (Lee, 2015) and the 2019 DEM is from USGS (refer to the Data Availability Statement for details; same for the other figures). Dashed and solid polygons delineate the margins of the old landslide deposits and the currently active landslide, respectively. Figure labels indicate the locations of features highlighted in subsequent figures. (b) Average GPS velocity (combined horizontal and vertical) during 7/98 to 7/99 (modified from Coe et al. [2000]).

2. Data and Methods

2.1. Elevation Data Sets

Two sets of freely available LiDAR DEMs were collected on 7 July 2015, and 26 August 2019 (refer to the Data Availability Statement for details). The National Center for Airborne Laser Mapping (NCALM) oversaw the LiDAR data collection in 2015 and the DEM raster has a resolution of 0.5 m and is archived at OpenTopography (Lee, 2015). The 2019 acquisition is part of the Southwest Colorado LiDAR acquisition task order issued by the USGS, and the DEM raster has a resolution of 1 m. For USGS's DEM mission, the reported root mean squared error (RMSE) in the vertical direction is less than 10 cm. Both DEMs apply the Universal Transverse Mercator (UTM) spatial reference system in conformance with the horizontal North American Datum of 1983 (NAD83). All bare-earth elevation estimates are in meters and are referenced to the North American Vertical Datum of 1988 (NAVD88). However, the NCALM mission applied Geoid 12A and the USGS mission applied Geoid 12B. We applied a constant adjustment based on the mean difference in their overlapping area (~14.81 m).

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2.2. Offset Tracking on DEM Shaded Relief Images for Horizontal Displacements

Sub-pixel offset tracking can be used to quantify surface displacements between images acquired at different times. The measured offsets are converted to distance by multiplying the pixel spacing by the sub-pixel (fractional) shifts, computed separately along image columns and rows, yielding two-dimensional displacement fields. Under favorable signal-to-noise conditions, the method can achieve accuracies of 1/10th–1/20th of a pixel (e.g., Hanssen, 2001). The method can be applied to optical imagery or SAR amplitude data. Because spaceborne optical and SAR amplitude pixels typically span a few to tens of meters, offset tracking is best suited for large displacements such as coseismic deformation and glacier movements (e.g., Hu et al., 2014; Michel et al., 1999; Scambos et al., 1992). Given the substantial movements of the Slumgullion landslide, offset tracking with remotely sensed images has also been used for analysis. For example, several studies used airborne UAVSAR images with a pixel spacing of 5.7 m in slant range and 7.2 m in azimuth to generate offset pairs with offset tracking (e.g., Delbridge et al., 2016; Hu, Bürgmann, Fielding, & Lee, 2020; Wang et al., 2016). Van Wyk de Vries et al. (2024) used 12 cloud-free, 10-m-spacing spaceborne Sentinel-2 L1C images collected between October 2016 and September 2022 to generate 49 offset pairs. Madson et al. (2019) previously applied offset tracking to shaded relief images derived from airborne LiDAR and UAV-based Structure-from-Motion products to measure Slumgullion landslide displacements. Here, we extended this approach by using two high-resolution, bare-earth LiDAR DEMs acquired four years apart, enabling spatially detailed displacement measurements and the reconstruction of 3D surface motion. At the Slumgullion landslide, the two high-resolution LiDAR DEMs acquired 4 years apart enable offset tracking to measure surface displacements at much finer spatial scales than in prior work. Offset tracking identifies the most likely match for a given target by searching for the peak of the cross-correlation within a moving window, so its performance depends on having clear surface texture. Several DEM secondary products (e.g., shaded relief, slope, aspect, curvature, and topographic wetness index) emphasize higher-frequency topographic variations relative to the raw elevations. Here we used the shaded relief because it preserves the two-dimensional texture needed for image matching. Shaded relief generation is controlled primarily by the illumination elevation and azimuth angles, which govern whether steep slopes are shadowed and the size and placement of those shadows. Standard default parameters are typically 40° (elevation) and 300° (azimuth). Using these settings, we observed that areas immediately downslope of the headscarps can fall into shadow, producing patches of zero values, although most surface textures remain well resolved. To reduce this effect, we additionally generated shaded relief images for both LiDAR DEMs using an alternative azimuth of 240° and compared the results (Figures S2 and S3 in Supporting Information S1 for details).

Point-cloud-based approaches provide an alternative for estimating three-dimensional surface motion; for example, Scott et al. (2020) applied a point-to-plane iterative closest point (ICP) algorithm to LiDAR data along the San Andreas Fault system. Compared with direct point-cloud methods, DEM-based offset tracking is computationally simpler and does not require access to the original point clouds, which may not always be available and often require additional expertise to process. More broadly, the method exploits the texture details of high-resolution DEMs and can therefore be applied to DEMs derived from LiDAR, photogrammetry, and ground-based InSAR.

Both DEMs were resampled and cropped to a common grid with 0.5 m pixel spacing. Window size strongly influences offset-tracking performance. If the window is too small, large pixel shifts may fall outside the search area, causing the algorithm to lock onto an incorrect cross-correlation peak and underestimate displacement. If the window is too large, precision can degrade and localized motion may be smoothed out, especially when most pixels within the window are stationary. The displacements across the landslide neck reach ~7 m per year (≈56 pixels over four years). Therefore, we used a 128×128 cross-correlation window and a 64×64-pixel search range.

2.3. Elevation Changes and Vertical Displacements

It is important to distinguish elevation change from vertical displacement. Elevation changes describe the difference in elevation at a fixed geographic location between two epochs, regardless of whether the surface material occupying that location has moved horizontally. Vertical displacement, in contrast, represents the change in elevation of the same ground target through time and therefore requires tracking that target between epochs. Accordingly, direct subtraction of co-registered DEMs yields elevation change at fixed coordinates and does not represent true vertical displacement where substantial horizontal motions occur. To estimate vertical displacement, we used the horizontal offsets derived from the DEM shaded relief images to back-translate each pixel in

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the 2019 DEM to its corresponding 2015 position. We then calculated the elevation difference between the back-translated 2019 DEM and the 2015 DEM. This difference represents the vertical displacement of the same tracked ground target rather than the elevation change at a fixed geographic location.

2.4. Horizontal Strain Rate

We first resampled the displacement velocities into 10-m grids. The horizontal strain-rate tensor was computed as the symmetric part of the horizontal velocity-gradient tensor (e.g., Bradley et al., 2019). Its normal components are $\dot{\varepsilon}_{ee} = \partial v_e/\partial e$ and $\dot{\varepsilon}_{nn} = \partial v_n/\partial n$, where v_e and v_n are the eastward and northward velocity components, respectively. Thus, $\dot{\varepsilon}_{ee}$ represents the rate of extension or contraction in the east–west direction, whereas $\dot{\varepsilon}_{nn}$ represents the rate of extension or contraction in the north–south direction; positive values indicate extension and negative values indicate contraction. The off-diagonal component, $\dot{\varepsilon}_{en} = 1/2(\partial v_e/\partial n + \partial v_n/\partial e)$, is the shear strain-rate component and describes angular distortion associated with spatial variations of the two horizontal velocity components. The principal strain rates, $\dot{\varepsilon}_1$ and $\dot{\varepsilon}_2$, are the maximum and minimum normal strain rates obtained as the eigenvalues of the strain-rate tensor:

$$\dot{\varepsilon}_{1,2} = \frac{\dot{\varepsilon}_{ee} + \dot{\varepsilon}_{nn}}{2} \pm \sqrt{\left(\frac{\dot{\varepsilon}_{ee} - \dot{\varepsilon}_{nn}}{2}\right)^2 + \dot{\varepsilon}_{en}{}^2}. \quad (1)$$

The maximum shear strain rate is:

$$\dot{\varepsilon}_{shear,max} = \frac{1}{2}\sqrt{(\dot{\varepsilon}_{ee} - \dot{\varepsilon}_{nn})^2 + 4\dot{\varepsilon}_{en}{}^2}. \quad (2)$$

All strain-rate quantities have units of inverse time, for example, yr^{-1} in this study.

Because strain rate is derived from spatial gradients of the velocity field, local estimates can be sensitive to measurement noise, outliers, and the spatial scale and method used to calculate the gradients, particularly where deformation is strongly heterogeneous (Pagani et al., 2021). Our comparison of strain rate results when applying sampling of displacement field show that the estimated strain rates are robust to reasonable variations in the sampling strategy, with no change in their order of magnitude (Figure S4 in Supporting Information S1). Consequently, local strain-rate extrema may be either overestimated or underestimated. We therefore interpret the absolute values together with their spatial distribution, placing greater confidence in coherent zones of extension, contraction, and shear than in isolated extreme values.

3. Results

3.1. Three-Dimensional Displacements From LiDAR DEM

We subtracted the elevations to determine the difference between the 2019 and 2015 DEMs (Figure 1). We delineated the external boundaries of the actively moving portion of the Slumgullion landslide using displacement data and topographic shaded relief images. Subsequently, we quantified elevation changes at the pixel level within the active landslide area. The area of elevation increase within the landslide is 1.56×10^6 m^2, corresponding to a volume gain of 3.90×10^5 m^3, whereas the area of elevation decrease is 1.64×10^6 m^2, corresponding to a volume loss of 4.10×10^5 m^3. Although the estimated volume gain and loss are of the same order of magnitude, a difference of approximately 2.0×10^4 m^3 (~5%) remains. This discrepancy may reflect uncertainties associated with DEM generation, co-registration, interpolation, and spatial coverage, as well as geomorphic processes such as localized erosion or deposition beyond the mapped landslide boundary. Therefore, the comparable magnitudes of volume gain and loss are consistent with, but do not by themselves demonstrate, mass conservation within the landslide.

The horizontal velocity field shows clear along-track variations: motions are slowest at the head, where sliding initiates; accelerate to a maximum at the neck where the channel is narrowest; and decrease to moderate and diverging rates at the toe, where the channel widens (Figures 2a and 2b). The horizontal displacements allow us to back-project the moving targets to their original horizontal position. We then obtained vertical movements associated with the three-dimensional landslide deformation by differencing the back-projected 2019 DEM and 2015 DEM.

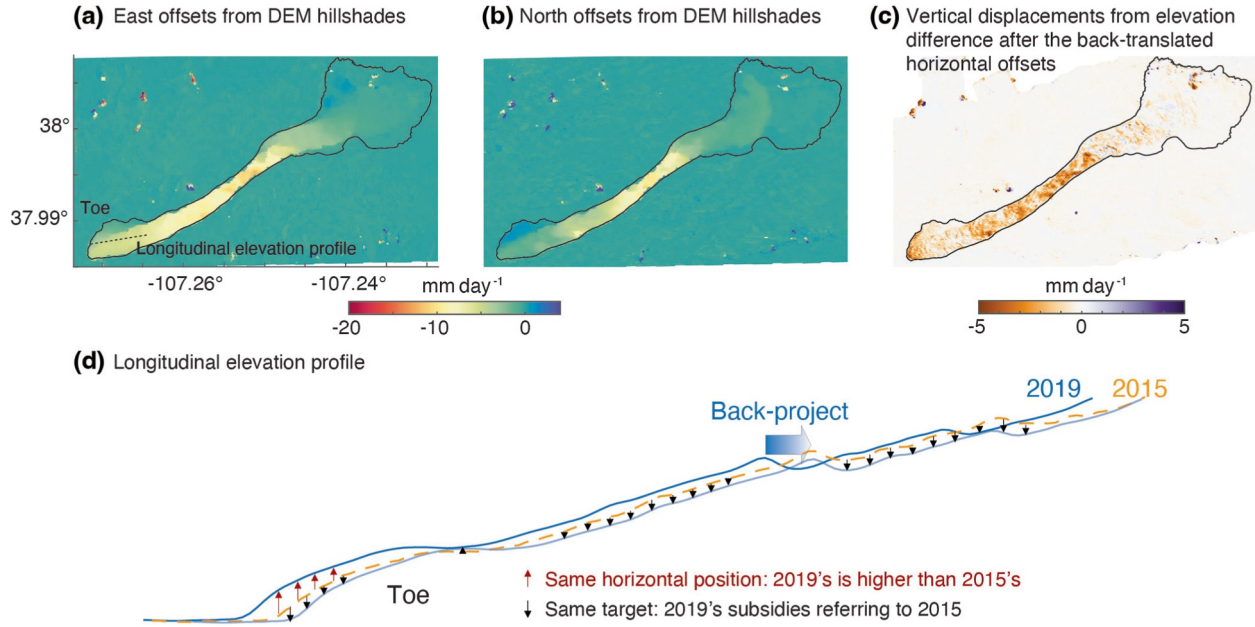


Figure 2. Three-dimensional displacements from the light detection and ranging digital elevation model (DEM). (a) and (b) East and north displacements computed from DEM shaded relief feature offsets. (c) Vertical displacements derived from the elevation difference after the back-translated horizontal offsets. (d) Sketch of the longitudinal elevation profiles (location shown by the dotted line in panel a) in 2015 (orange dashed line) and 2019 (blue solid line) over the toe, demonstrating the disparity between the elevation changes (upward red arrows) and vertical displacements (mostly downward black arrows).

The landslide toe shows an apparent elevation increase and exhibits vertical subsidence, which at first seems contradictory (Figures 1 and 2c). However, this behavior is consistent with the geometry illustrated in the sketch in Figure 2d. The longitudinal elevation profile in 2019 (blue line) is displaced primarily horizontally relative to its 2015 position, so elevations at the toe in 2019 are higher than in 2015 simply because the toe has advanced, resulting in higher topography (red arrows pointing up). When the motion is decomposed into horizontal and vertical components and the 2019 profile is shifted back horizontally to track the same targets, the inferred vertical component indicates downward motion (black arrows pointing down).

3.2. Secondary Mini-Slides

Secondary mini-slides can occur within the Slumgullion landslide. Coe et al. (2003) reported that intense precipitation in July 1999 triggered several debris flows on the Slumgullion landslide. However, information about the number, distribution and kinematics of the mini-slides is still limited. The only mini-slide that has been detailed is located in the lower part of the Slumgullion landslide (Figure 3) (Delbridge et al., 2016; Hu, Bürgmann, Fielding, & Lee, 2020).

Pull-apart basins are formed by the tension and depression in between sub-parallel strike slip faults. One such feature has been established as a part of this previously identified mini-slide (Hu, Bürgmann, Fielding, & Lee, 2020). Shaded relief images from 2015 to 2019 LiDAR DEMs show that the pull-apart basin has shifted downhill by ∼14 m during 2015–2019 at a rate of 3.5 m yr^{-1} (orange and blue polygons in Figure 3). Additionally, the cross-landslide-fault gullies shifted by ∼10 m at a rate of 2.5 m yr^{-1}. This offset deviates slightly from the previous lateral margins mapped from the landslide fault lineation, probably due to hummocky topography. The distance between the pull-apart basin and the cross-landslide-fault gullies is ∼60 m, and the landslide rates decrease from 3.5 to 2.5 m yr^{-1} and diminish to zero immediately, indicating the lateral shearing surfaces are very steep. The stream-channel offsets are similar to the well-known Wallace Creek and Strawberry Creek in California (Zielke et al., 2010).

In addition to the secondary mini-slide against the southern margins in the lower part of the landslide, the differential DEM map allows us to identify two additional mini-slides not reported before, one below the middle part of the headscarp (Figure 4), and the other in the big bend zone of stretching in the upper part of the landslide (Figure 5).

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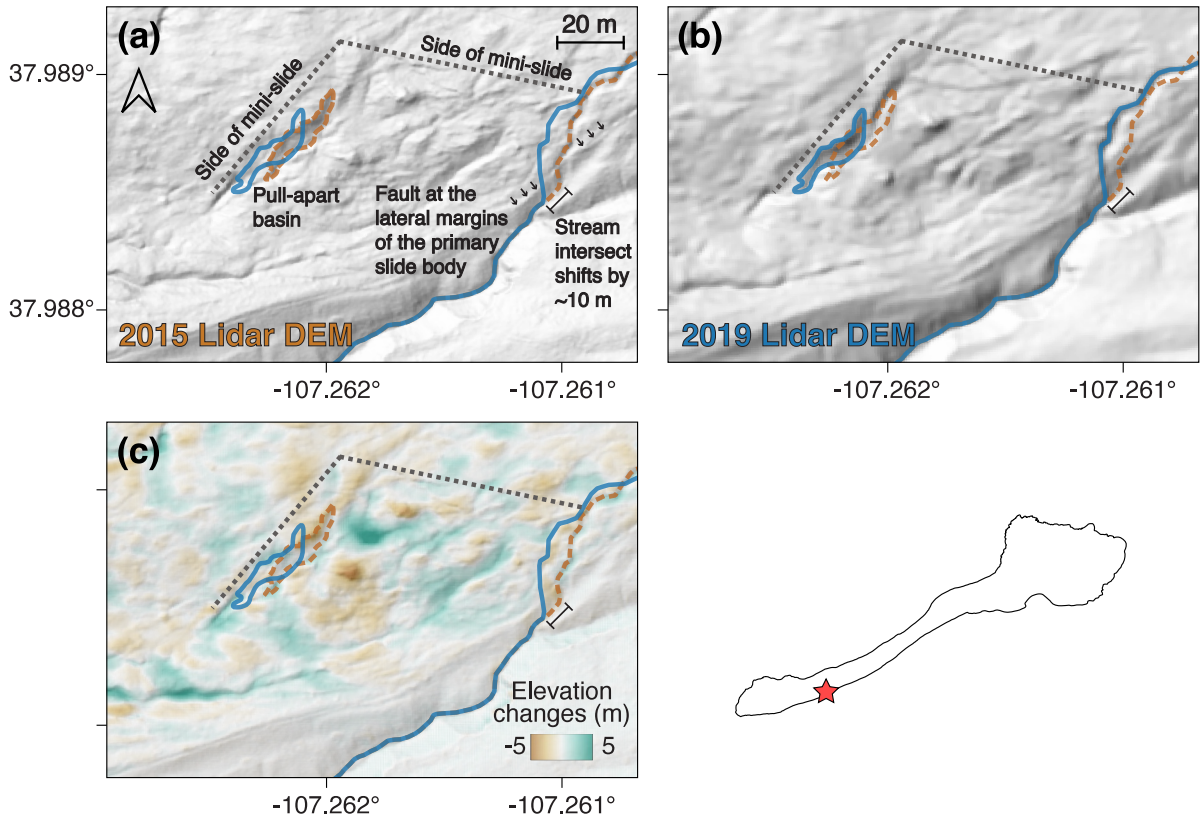


Figure 3. Older, secondary mini-slide in the lower part of the landslide against the southern margins. (a, b) The shaded relief images of light detection and ranging digital elevation models collected in 2015 and 2019, respectively. (c) The elevation changes. Black dashed lines show two sides of the mini-slide moving toward the landslide fault at the lateral margins of the primary slide body. Orange and blue polygons show the position of the pull-apart basin in 2015 and 2019, respectively. Similarly, orange and blue lines show the positions of the stream intersect by the landslide external margins.

One of the additional identified mini-slides is located near the landslide headscarp (Figure 4). The near-vertical main headscarp is characterized by hydrothermally altered Tertiary volcanic rocks. Morphological modifications of the main scarp, avalanche chutes, and associated talus deposits have been documented through terrestrial photographic surveys. Schuster and Smith (1996) attributed the progressive retreat of the headscarp to the combined effects of freeze-thaw weathering, sapping, and erosional processes. The western section of the landslide's head moves at a moderate rate, sliding against a curved flank, separated from the eastern section that moves almost undetectably. The eastern active section was missed in the original version of the kinematic elements mapped in 1991 (Parise & Guzzi, 1992) but was identified by Hu, Bürgmann, Schulz, and Fielding (2020) from InSAR analyses. A lobe below the headscarp is actively sliding as an independent mini-slide, where its upper part is depleting (elevation decrease), transporting along a curved path to the accumulation zone (elevation increase); the total travel path is over 300 m (Figure 4). This mini-slide is steep with an average slope of ~34°. Our findings indicate that the landslide head continues to exhibit active movement. The load generated by head collapse is believed to have served as the trigger for the landslide reactivation ~300 years ago (e.g., Fleming et al., 1999).

The other newly discovered mini-slide is identified at the beginning of the zone of stretching outside of a bending ridge against the southern margin of the landslide. Beyond the previously mapped external margins marked by a relatively deep-incised stream, this mini-slide tends to move northwesterly into the main landslide body (Figure 5). The boundary between the areas of depletion and accumulation can be captured from the distinct transition between the positive and negative elevation changes (Figure 5).

Swarms of fracture zones have formed in the zone of stretching between the landslide head and neck. These fractures can help promote water infiltration from the surface (Coe et al., 2003). The direction of the linear furrows is orthogonal to the longitudinal axis (AB in Figure 6). This area is dominated by extension driven by

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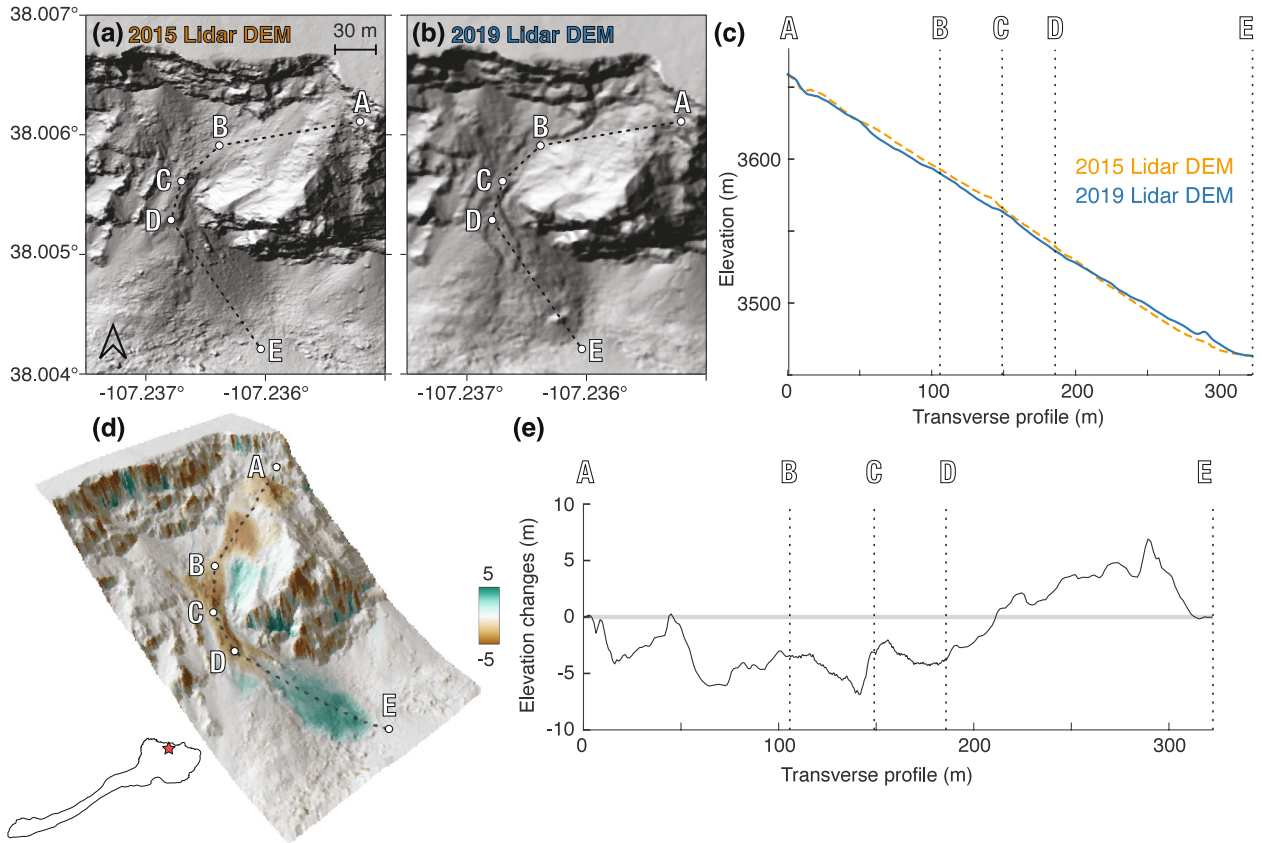


Figure 4. Newly discovered secondary mini-slide below the middle part of the headscarp. (a, b) The shaded relief maps of light detection and ranging digital elevation models collected in 2015 and 2019, respectively. (c) The elevation profile along a curved path denoted by A-B-C-D-E in panels (a, b, and d). (d) The elevation changes are superimposed on a 3D map. (e) The elevation change profile shows a decline on the head and increase on the toe at this mini-slide.

differential movement: the lower slope moved ~16.6 m downhill over 4 years, while the upper slope moved only ~10.7 m. The mass has been pulled apart and formed fractures, cracks, and normal faults along weak planes. An overall mass thinning is expected in such an extensional environment. Tilted blocks have created the miniature of northwest-trending basins and ranges in this area. By shifting the longitudinal elevation profile in 2015 by ~13 m downhill (direction of A pointing to B in Figure 6), the 1st-order variations of the elevation profiles from 2015 to 2019 align closely, indicating a translational slide and generally smooth basal surfaces at the entrance of landslide neck; otherwise, substantial surface uplift or subsidence would be apparent.

4. Discussion

4.1. Decelerating Movement of the Overriding Toe

Considering projected changes in the moisture-balance drought index, Coe (2012) suggested that the landslide head might cease moving by the mid-21st century. Because the head collapse has been argued to act as the trigger for ongoing activity (Fleming et al., 1999), Coe (2012) further inferred that landslide velocities could decline under future warming climates. Recall that our detailed LiDAR mapping indicates a small portion of the landslide head is actively collapsing (Figure 4); at least for now, the landslide continues to be supplied with new materials. We also collected the precipitation, air temperature, and soil moisture data reporting since the 1980s from SNOwpack TELemetry Network (SNOTEL) site 762 (refer to the Data Availability Statement for details), ~4 km southeast of the head of the landslide. The amount of precipitation remains relatively stable, yet individual wet and dry years can still be identified.

Although previous studies have proposed that climate change may influence landslide activity, the observational record has not yet clearly documented how Slumgullion's displacement rates have changed over decadal

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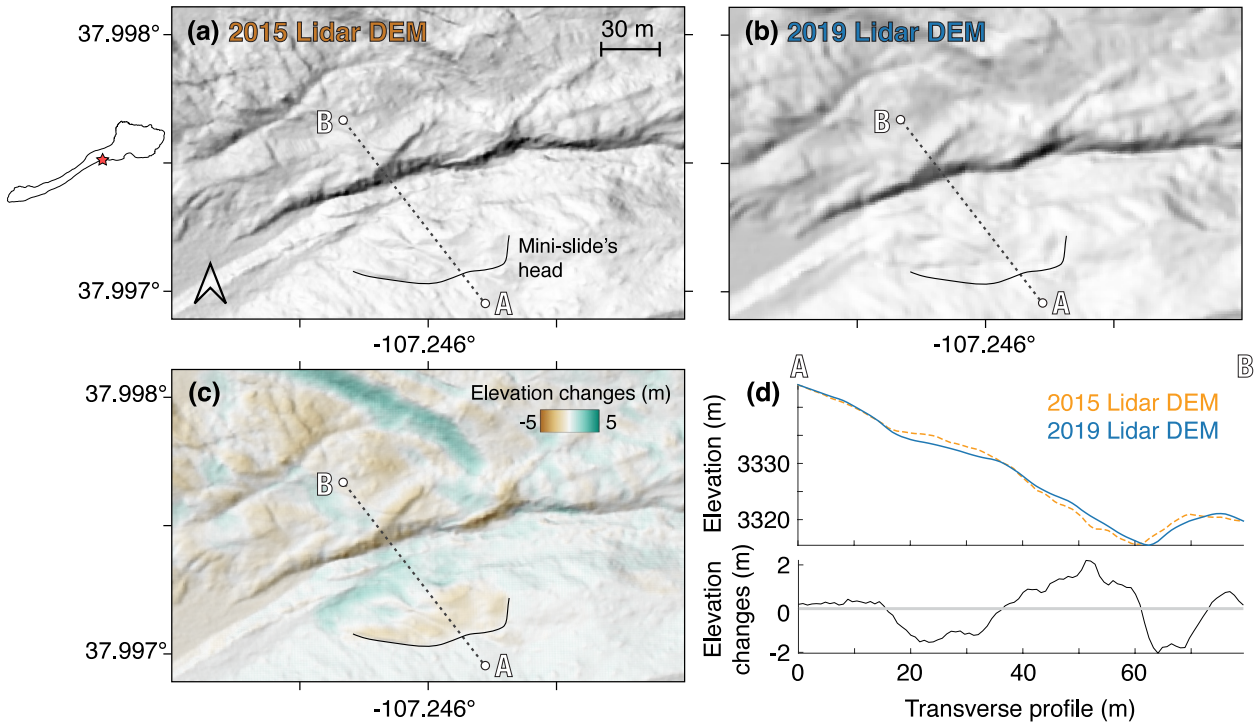


Figure 5. Newly discovered secondary mini-slide bounding the zone of stretching in the upper part of the landslide against the southern margins. (a, b) The shaded relief images of light detection and ranging digital elevation models collected in 2015 and 2019, respectively. (c) The elevation changes. (d) The elevation profile along A–B in panels (a–c).

timescales. To investigate longer-term kinematic changes, we used the advancing landslide toe as a representative area and synthesized historical measurements with our recent observations. We found that the longitudinal surface profiles have been translationally sliding downhill. Obstacles such as rocks and trees bounding the toe have been damaged or engulfed by the landslide mass (Hu & Bürgmann, 2020). Therefore, the active landslide keeps growing in length. The off-toe slope is relatively flat (6°). The advancing movements almost entirely rely on pushing from the translating slide mass. This creates a steepened wall (>60°) more than 2 m high at the frontal edge of the toe. Next, we synthesized the toe displacement results from previous studies, presented our more recent observations, and linked these episodes into a continuous record as described below.

Revisiting the observational record through time, Crandell and Varnes (1961) estimated the long-term toe velocities of ∼0.75 m yr^{-1} using air photos collected during 1940–1952. Jackson et al. (1996) reported an average toe velocity of ∼0.4 cm day^{-1} (≈1.46 m yr^{-1}) based on traditional control surveys spanning roughly the 35 years prior to their publication, that is, since the ∼1960s. Smith (1993) used aerial photographs from 1985 to 1990 (dates not reported in the original study; the Bulletin 2130 (Varnes & Savage, 1996) summary lists 8/12/1985 and 8/25/1990) to map displacements at 310 slide points. Although toe motion was not measured across the ground surface, Smith quantified displacement near the toe crest, finding a total shift of ∼9.5–9.8 m over 5 years, equivalent to ∼1.9–2.0 m yr^{-1}. Fleming et al. (1999) reported stake-based point displacements from survey measurements on and off the toe between September 1992 and September 1993, indicating a velocity of at least ∼1.93 m yr^{-1}. Savage and Fleming (1996) deployed a creepmeter at the toe from April 9 to November 9,1993. Their measurements indicate a brief episode of rapid motion (∼5.2 m yr^{-1}) from April 9 to May 18, followed by substantially slower displacement (∼1.0 m yr^{-1}) from May 18 to November 9. Despite data gaps, the creepmeter record (covering ∼13 of 21 weeks) indicated relatively steady rates of ∼1.40 m yr^{-1} from July 13 to September 15 and ∼1.46 m yr^{-1} from September 15 to 9 November 1993. Using stereo aerial photographs from 1939, 1940, and 2000, Coe et al. (2009) found that the toe advanced ∼92 m over 1939–2000, corresponding to an average rate of ∼1.5 m yr^{-1}. Hu and Bürgmann (2020) reported an average toe movement rate of ∼1.5 m yr^{-1} when comparing UAVSAR observations from 2011 to 2018 with a 1991 field survey (Figure 7).

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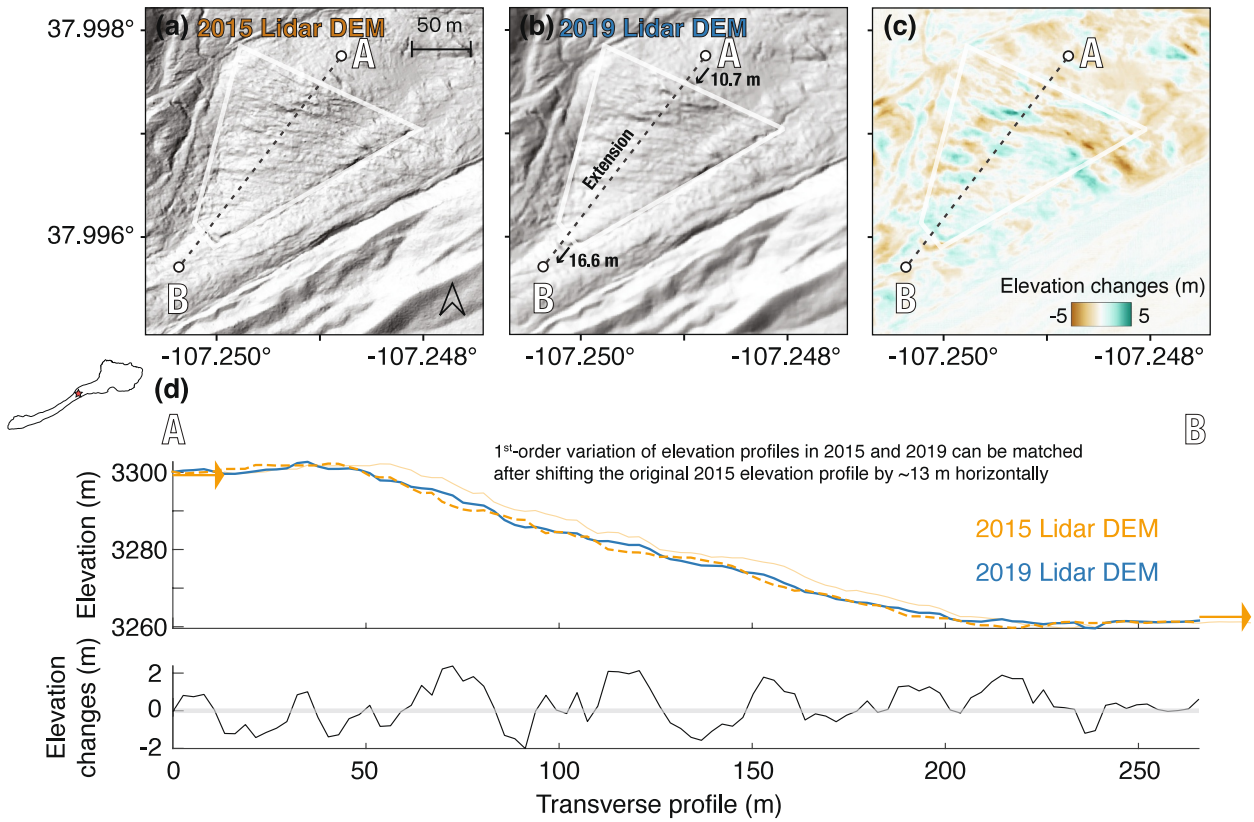


Figure 6. Extension along the longitudinal axis in the zone of stretching. (a, b) The shaded relief images of light detection and ranging digital elevation models collected in 2015 and 2019, respectively. (c) The elevation changes. (d) The profiles of elevation and elevation changes along A–B in panels (a–c). The orange dashed and blue solid lines show the elevation profiles in 2015 and 2019, respectively. The light-orange solid line represents the 2015 elevation profile after applying a horizontal shift to align its topographic features with those observed in 2019, indicating predominantly translational motion.

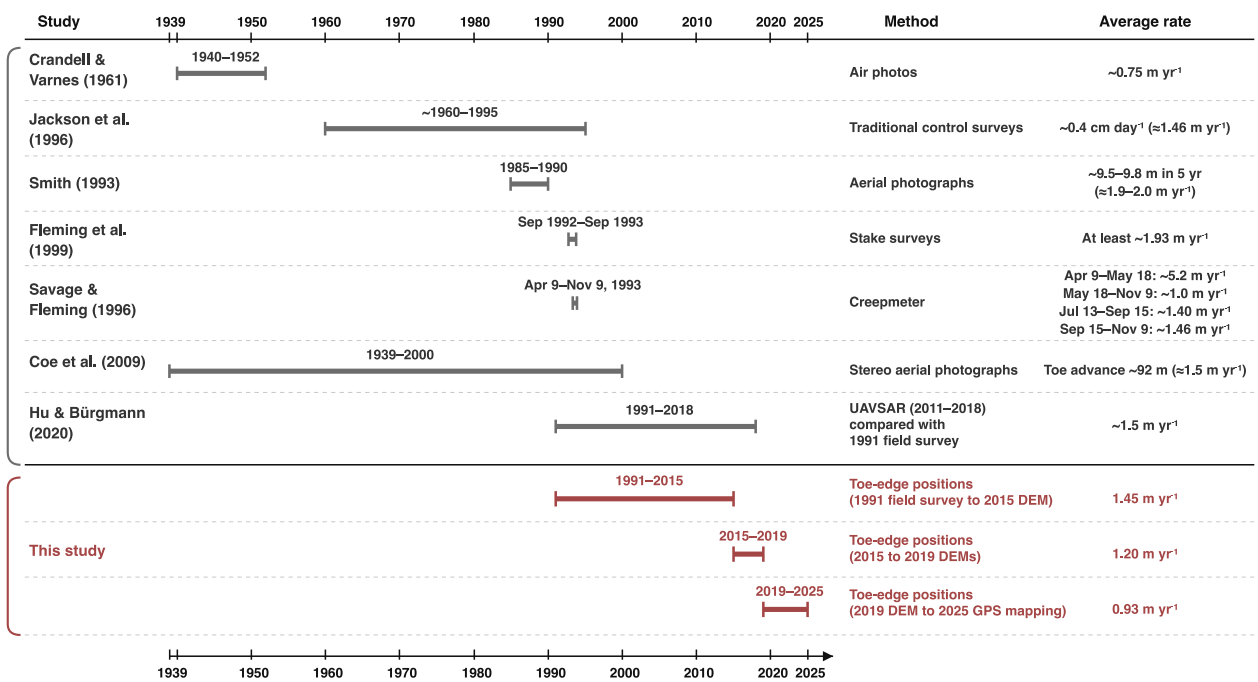


Figure 7. Timeline of previous displacement velocity measurements over Slumgullion landslide's toe and results from this study.

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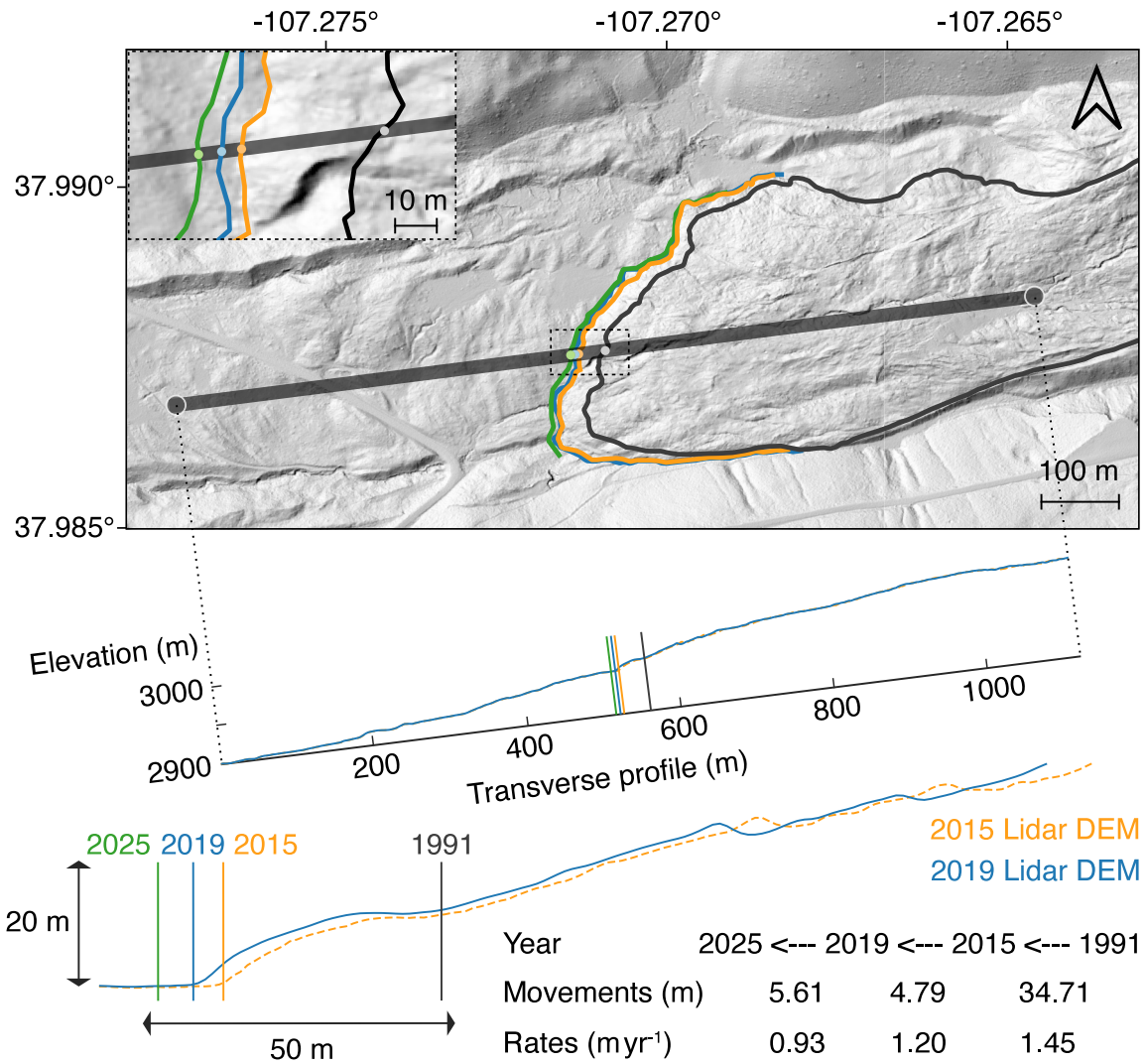


Figure 8. The geographic position of toe edges in 1991 (gray), 2015 (orange), 2019 (blue), and 2025 (green).

The historical records are not always consistent, likely because (a) different methods may measure motion along different orientations (not necessarily the downslope direction) and carry different uncertainties; (b) different parts of the toe move at different rates and measurement locations were not always documented; (c) some early measurements span only short intervals, and the timing within the year matters because landslide motion is strongly seasonal.

We obtained the geographic locations of the toe edges in 1991, 2015, 2019, and 2025, spanning 34 years (Figure 8). The earliest version was derived from the field survey in 1991 (Parise & Guzzi, 1992). The 2015 and 2019 versions were extracted from the DEM where the toe sits atop the overrun ground surface. The 2025 version was mapped using a portable GPS receiver at the landslide site. Taking the toe edge as the reference, we measured its movements between the time epochs and computed the annual rate. Our results show that the landslide rates decelerated from 1.45 m yr^{-1} during 1991–2015, to 1.20 m yr^{-1} during 2015-2019, and to 0.93 m yr^{-1} during 2019–2025. Even when incorporating the historical records, excluding the Crandell and Varnes (1961) estimate, which is much smaller than other values (including Coe et al., 2009 that covered part of the same time frame), the conclusion that the Slumgullion landslide has been decelerating still holds.

Relative to the historical movement rate of approximately 1.50 m yr^{-1} during 1939–2000 reported by Coe et al. (2009), the present-day rate of 0.93 m yr^{-1} represents a reduction of approximately 38% [((1.50–0.93)/1.50) × 100%]. Likewise, compared with the average rate of 1.45 m yr^{-1} during 1991–2015, the current rate is

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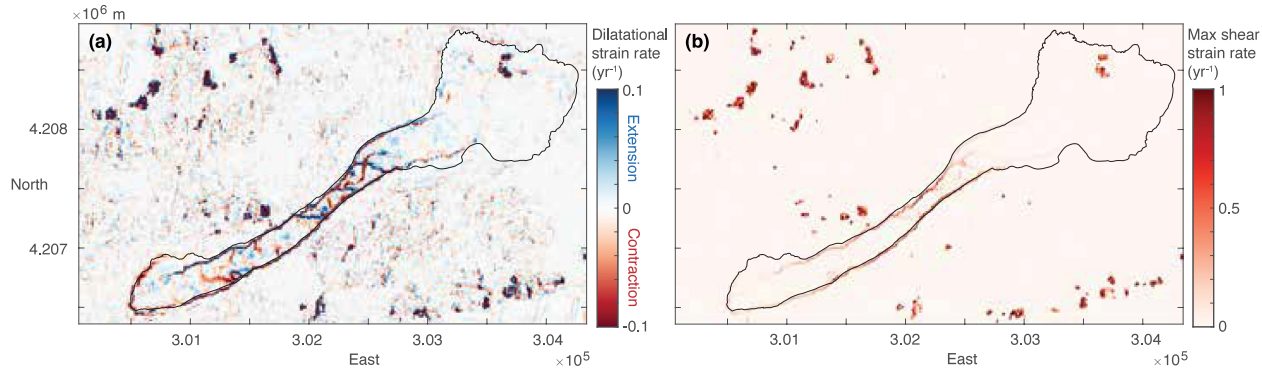


Figure 9. Horizontal strain rates of the Slumgullion landslide. (a) Areal strain rate (dilatation). Positive (bluish) values indicate extension and negative (reddish) values indicate contraction. (b) Maximum shear strain rate.

approximately 36% lower [((1.45–0.93)/1.45) × 100%]. Therefore, most of the rate changes occurred during more recent decades. By leveraging the reported observations and our results, our results demonstrate that landslide deceleration has become more severe in the past decades. In this scenario, the landslide potentially could deactivate over the coming decades.

4.2. Spatial Distribution of Strain Rate

Landslides and tectonic faults share similar deformation styles, ranging from slow creep to rapid failure. Their characteristic spatial and temporal scales, however, differ substantially. Quantifying strain rates at Slumgullion landslide therefore provides an opportunity not only to characterize internal landslide deformation, but also to examine fault-like deformation processes occurring over reduced spatial dimensions and accelerated timescales (e.g., Shi & Hu, 2023).

At Slumgullion landslide, horizontal areal strain rates (dilatation) reach ∼0.4 yr^{-1} within the landslide interior (Figure 9a), and the maximum shear strain rates approach ∼1 yr^{-1} along the external margins (Figure 9b), indicating intense shearing. The toe exhibits clear contraction, and the landslide neck and secondary mini-slides contain localized patches of contraction and extension. These spatial patterns reveal internal kinematic structures and help refine the boundaries of kinematic elements.

Strain rates reported for other landslides are substantially lower, despite comparable spatial sampling of the horizontal deformation fields. The Mud Creek landslide (California, USA) failed catastrophically on 20 May 2017, destroying California State Highway 1; the strain rate was on the order of $\sim10^{-3}$ yr^{-1} inferred from InSAR-derived horizontal mean velocities from 2009 to 2017 (Handwerger et al., 2019, 2025). The Hooskanaden landslide (Oregon, USA) failed in 1977, 1995, and 2019; InSAR mapping from 2016 to 2019 prior to the most recent failure indicates horizontal strain rate on the order of $\sim10^{-3}$ yr^{-1} (Zheng et al., 2024). For episodic events, strain instead of strain rate is usually computed. The 2018 M7.5 Sulawesi earthquake triggered widespread landslides in Palu Valley in Indonesia; horizontal displacements estimated via pixel-level correlation of pre- and post-event Planet imagery yielded strains on the order of $\sim10^{-2}$ (Bradley et al., 2019).

The contrast is distinct when compared with tectonic deformation. Interseismic strain rates span from $\sim10^{-9}$–10^{-8} yr^{-1} in stable plate interiors to $\sim10^{-7}$–10^{-6} yr^{-1} in active plate-boundary zones and across locked faults (Shen & Liu, 2025; Wright et al., 2026). For dynamic earthquakes, the event strain inferred from geodetic displacement gradients typically spans $\sim10^{-9}$–10^{-7} in the far field, increases to $\sim10^{-5}$–10^{-3} at regional to near-fault distances where strain fields are mapped directly from co-seismic displacement products, and can reach $\sim10^{-2}$ or higher where deformation localizes in damage zones or ground failures, for example, large near-rupture shear strains mapped for 2019 Ridgecrest earthquake (Milliner et al., 2021).

Thus, the strain rates at Slumgullion landslide are orders of magnitude higher than those typically observed in other slow-moving landslides and during interseismic tectonic deformation. Importantly, these high rates occur within structures that resemble tectonic deformation systems: shearing along landslide boundaries has produced strike-slip faults along its lateral margins, and distributed deformation includes pull-apart basins within the

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dilatational shear zones and transverse ridges where displacement vectors rotate along the curved channel (Schulz, McKenna, et al., 2009). Slumgullion landslide therefore provides a natural laboratory to investigate failure processes that evolve much more slowly in tectonic systems.

5. Concluding Remarks

Generations of multi-source data collection and research on the Slumgullion landslide have enabled advancements in data processing techniques and improved understanding of landslide mechanisms. In this study, we first calculated horizontal displacements by measuring pixel offsets between shaded relief images derived from high-resolution LiDAR DEMs acquired 4 years apart. We represented the application of offset-tracking method using DEM-derived shaded relief images as the data source, and this approach can be readily extended to other DEM derivatives such as slope, aspect, and curvature, and to other settings such as earthquakes and glaciers. The highly accurate LiDAR DEMs also allow us to extract elevation changes and vertical displacements. Second, in addition to the previously reported secondary mini-slide along the southern margin in the lower portion of the landslide, we identified new secondary failures, including one at the headscarp. This newly collapsed part of the head contributes new materials that continue to fuel the present-day downslope slide movement; however, the toe continues to decelerate. Third, to analyze long-term landslide rate change, we relied on multi-source and multi-temporal data, combined with previous field surveys and our recent surveys, to reveal a general trend of deceleration in landslide movement since 1939. Overall, this study enhances understanding of the Slumgullion landslide's long-term movement and provides high-resolution 3D displacement field to support future in situ sensor placement and kinematic analysis. Legacy Slumgullion landslide records, many preserved as hard copies, together with more recent aerial images and data from advanced sensors from multiple laboratories indicate that observations of the landslide have been accumulating for decades or even centuries. A dedicated effort to synthesize and integrate these valuable data sets could enable a more comprehensive understanding of this globally recognized natural laboratory.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

LiDAR DEM was issued by the U.S. Geological Survey (2022). LiDAR DEM collected by National Center for Airborne Laser Mapping (NCALM) is distributed by OpenTopography (Lee, 2015). The meteorological data collected at the SNOwpack TELemetry Network (SNOTEL) site 762 are available from the Natural Resources Conservation Service of the U.S. Department of Agriculture (2026).

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